



NITROXHIFT STUDIOS

Web Development Internship 2026

DIGITAL SKILLS PLATFORM

Instructor Dashboard

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WEB DEVELOPMENT INTERNSHIP

FINAL TASK

Project: Digital Skills Platform

Module: Instructor Dashboard

Frontend: React.js + Vite

Backend: FastAPI

Database: PostgreSQL

API: REST API

Digital Skills Platform

Project Title

Digital Skills Platform - Instructor Dashboard

Introduction

The **Digital Skills Platform – Instructor Dashboard** is a full-stack web application developed as an individual Week contribution for **Nitro Solutions**. The dashboard is designed to help instructors manage courses, view students, track student progress, and manage grades.

The application uses **React.js** for the frontend, **FastAPI** for the backend, **PostgreSQL** for the database, and **REST APIs** for communication between the frontend and backend.

The project demonstrates practical skills in full-stack web development, API development, database management, responsive UI design, and Git/GitHub version control.

Project Overview

The Digital Skills Platform – Instructor Dashboard was developed as an individual Week contribution for Nitro Solutions. The main purpose of this module is to provide instructors with an easy-to-use interface for managing courses, viewing students, tracking student progress, and managing grades.

The project is a full-stack web application developed using React.js, FastAPI, PostgreSQL, and REST APIs.

Project Objective

The objective of this project is to build an Instructor Dashboard that allows instructors to:

- Create, edit, and delete courses
- View registered students
- View and update student progress
- Add, view, and delete student grades
- Login and manage their instructor account

Technologies Used

Frontend: React.js, Vite, JavaScript, HTML5, CSS3

Backend: Python, FastAPI, Pydantic, SQLAlchemy

Database: PostgreSQL

Tools: Visual Studio Code, Git, GitHub, Swagger UI, pgAdmin

How the System Works

The application follows a simple full-stack architecture:

React Frontend



REST API



FastAPI Backend



PostgreSQL Database

The instructor interacts with the React dashboard. The frontend sends requests to the FastAPI REST API. FastAPI processes the requests and stores or retrieves data from PostgreSQL using SQLAlchemy.

Main Features

Authentication

- Student and Instructor registration
- Login functionality
- Role-based dashboard
- Logout

Course Management

- Create courses
- View courses
- Edit courses
- Delete courses

Student Progress

- View student progress
- Update progress from 0% to 100%

Grade Management

- Add student grades
- View grades
- Delete grades
- Marks validation from 0 to 100

Database Management

The PostgreSQL database contains the following main tables:

- Users
- Courses
- Enrollments

- Progress
- Grades

Challenges and Solutions

During development, an **Unknown Course** issue occurred because some progress records referred to courses that had already been deleted. This was solved by retrieving course information using **course_id** and deleting related progress, grades, and enrollment records when a course is deleted.

Another challenge was communication between the React frontend and FastAPI backend. This was resolved by configuring CORS in FastAPI.

How to Run the Project

Backend

```
cd backend
```

```
uvicorn main:app --reload
```

Backend:

<http://127.0.0.1:8000>

Swagger:

<http://127.0.0.1:8000/docs>

Frontend

```
cd frontend
```

```
npm install
```

```
npm run dev
```

Frontend:

<http://localhost:5173>

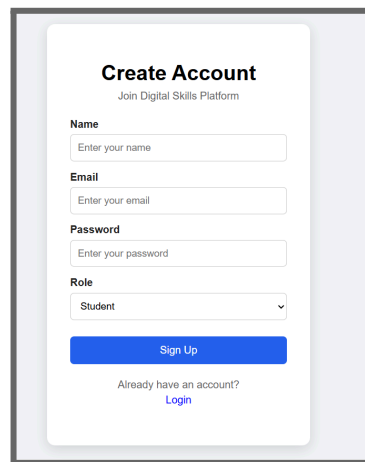
Testing

The following features were tested successfully:

- Registration and Login
- Instructor Dashboard
- Course CRUD operations
- Student management
- Progress management
- Grade management
- PostgreSQL database operations
- REST APIs through Swagger

Screenshots

Figure 1: SignupPage



The screenshot shows a 'Create Account' form for a 'Join Digital Skills Platform'. The form includes input fields for 'Name' (placeholder: 'Enter your name'), 'Email' (placeholder: 'Enter your email'), and 'Password' (placeholder: 'Enter your password'). There is a 'Role' dropdown menu with 'Student' selected. A blue 'Sign Up' button is at the bottom. Below the button, it says 'Already have an account?' with a blue 'Login' link.

Figure 2: Login Page

Login

Welcome back to Digital Skills Platform

Email

Password

Login

Create a new account

Figure 3: Instructor Dashboard

Instructor Panel

Dashboard

Courses

Students

Progress

Grading

Help

Welcome, Instructor daniel

Courses

Add, edit and manage your courses

View Courses

Students

View and manage students

View Students

Progress

Check student course progress

View Progress

Grading

Add and manage student grades

View Grades

Figure 4: Course Management

Manage Courses

Add new courses or update existing courses.

Add New Course

Course Title

Course Description

Course Content

Add Course

AI

Create models

Content

Python

View

Edit

Web development

Create websites

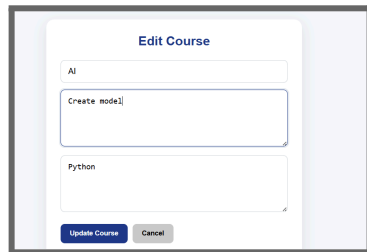
Content

HTML, CSS, JS & React

View

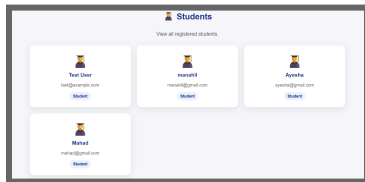
Edit

Figure 5: Edit Course



The 'Edit Course' form contains three input fields: a text field with 'AI', a text area with 'Create mode()', and a text field with 'Python'. At the bottom are 'Update Course' and 'Cancel' buttons.

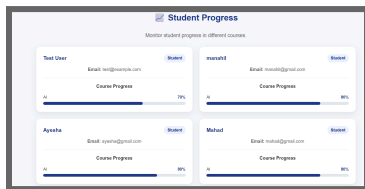
Figure 6: Students



The 'Students' page shows a list of four registered students, each with a profile card containing an icon, name, email, and a 'Student' link.

Name	Email
Test User	test@exampl.com
muhammad	muhammad@gmail.com
Ayasha	ayasha@gmail.com
Muhammad	muhammad@gmail.com

Figure 7: Student Progress



The 'Student Progress' page displays progress bars for four students across two courses (AI and Python). The progress is 100% for all.

Student	AI Progress	Python Progress
Test User	100%	100%
muhammad	100%	100%
Ayasha	100%	100%
Muhammad	100%	100%

Figure 8: Grade Management

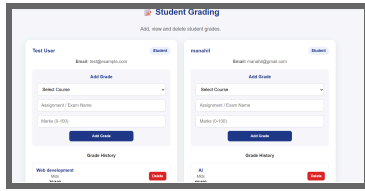
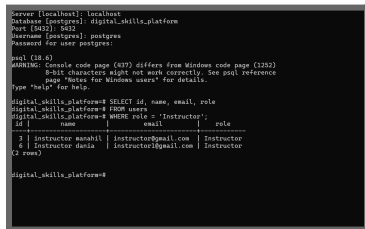


Figure 9: Swagger API



Figure 10: PostgreSQL



Conclusion

The Instructor Dashboard successfully implements the main requirements of the Week contribution. It provides instructors with a centralized system for managing courses, students, progress, and grades.

This project provided practical experience in full-stack web development, REST API development, PostgreSQL database management, responsive UI design, debugging, and Git/GitHub version control.

Progress Report

Task	Status
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React/Vite Setup	Completed
Instructor Dashboard	Completed
Authentication	Completed
Course Management	Completed
Add Course	Completed
Edit Course	Completed
Delete Course	Completed
Student Management	Completed
Student Progress	Completed
Grade Management	Completed
Grade Delete	Completed
FastAPI REST APIs	Completed
PostgreSQL Database	Completed
Responsive UI	Completed
API Testing with Swagger	Completed
GitHub Repository	Completed
README	Completed

Screenshots	Completed
Documentation	Completed
Final Working Demo	Completed

Explanation Video	Completed
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